

CMS Workflow Management Requirements

A. Malta Rodrigues and A. Piccinelli (U. of Notre Dame)
on behalf of the CMS Offline & Computing Coordination Area

Acknowledgments to the stakeholders who contributed

Core software

Matti Kortelainen, Chris Jones

Data Management

Hasan Ozturk, Panos Paparrigopoulos, Eric Vandeering

DBS and DAS

Valentin Kuznetsov, Todor Ivanov, Stefano Belforte

Generation group

Konstantinos Theofilatos, Lorenzo Viliari

P&R

Jen Adelman-McCarthy, Muhammad Hassan Ahmed, Kirill Ivanov, Dmytro Kovalskyi

Physics coordinators

Guillermo Gomez-Ceballos, Andreas Meyer

PPD

Jean-Roch Vlimant

RECO

Andrea Bocci, Charis Kleio Koraka, Mohamed Darwish

Submission infrastructure

Marco Mascheroni, Antonio Perez, Todor Ivanov

Tier-0

Jan Eysermans, Antonio Linares, Ohm

User/private workflow support

Stefano Belforte, Vijay Chakravarty, Katy Ellis

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Editing policy and scope conventions

- Preserve CMS-specific language and stakeholder intent where possible, while generalizing implementation-specific wording.
- Avoid current-system component names and the term "agent" unless explicitly discussing legacy migration or historical context.
- Separate requirement criticality from implementation ownership. A capability can be hard but still depend on another CMS system.
- Requirements that are not WMS-owned are retained as external dependencies, interface constraints, or policy decisions rather than silently discarded.
- Runtime-generated globally unique output identifiers are taken as the preferred direction for output identity across retries and recovery.

Requirement levels

Hard requirement: The new WMS is unlikely to be viable without satisfying this capability.

Soft requirement: A desired capability that improves HL-LHC readiness, scalability, usability, or maintainability, but may be phased or scoped.

Open decision: A capability or design point that depends on a CMS policy, architecture, ownership, or stakeholder decision that is not yet settled.

Scope / likely owner

WMS core: Expected to be implemented primarily by the WMS architecture and services.

WMS interface / dependency: The WMS must consume, expose, propagate, or remain compatible with a capability owned partly or fully elsewhere.

Co-owned WMS + external systems: Requires joint design across WMS and another CMS system or stakeholder area.

External dependency / policy decision: Important for the future computing model, but not a WMS-owned implementation requirement.

Design principle: Guidance for architecture choices rather than a testable numbered requirement.

Definition priorities

Definition priority indicates how early a requirement or open decision must be clarified because it constrains the architecture, interfaces, or sequencing of other requirements. It is not the same as requirement level: a hard requirement may have a later definition priority if it does not block early architectural choices, while a soft or open item may have high definition priority if it affects many downstream decisions.

P0 — Blocking design decision: Must be clarified early because it constrains the overall WMS architecture, major interfaces, ownership boundaries, or evaluation of candidate solutions.

P1 — Early architecture requirement: Should be defined during the early design phase because it affects important interfaces, service boundaries, data flows, or operational models, but does not block every other architectural choice.

P2 — Detailed design / phased capability: Can be specified later once the main architecture and ownership model are clearer. These items may still be important, but they are less likely to block the initial architecture definition.

Requirement categorization summary

This draft contains 69 numbered CMS-WM requirements. Requirement level and definition priority serve different purposes: the requirement level describes the criticality of the capability, while the definition priority indicates how early the requirement or associated open decision must be clarified for architecture planning.

Global categorization matrix (counts)

Level \ Definition priority	P0	P1	P2	Total
Hard requirement	12	45	2	59
Soft requirement	1	3	4	8
Open decision	2	0	0	2
Total	15	48	6	69

P0 requirements: CMS-WM-002, CMS-WM-004, CMS-WM-008, CMS-WM-013, CMS-WM-018, CMS-WM-026, CMS-WM-029, CMS-WM-030, CMS-WM-036, CMS-WM-043, CMS-WM-046, CMS-WM-047, CMS-WM-060, CMS-WM-061, CMS-WM-065.

The section-level matrices below list the specific requirement IDs falling into each Level / Definition priority category; the requirement text remains unchanged.

1. Workflow model, validation, and controlled replay

Section categorization matrix (requirement IDs)

Level \ Definition priority	P0	P1	P2	Total
Hard requirement	CMS-WM-002, CMS-WM-004	CMS-WM-001, CMS-WM-003, CMS-WM-005, CMS-WM-006, CMS-WM-007	—	7
Soft requirement	—	—	—	0
Open decision	—	—	—	0
Total	2	5	0	7

CMS-WM-001 - Complete native workflow state transitions

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS core

Supporting groups: P&R, PPD

Requirement: The new WMS must be a complete system capable of handling workflow state transitions internally. External operational layers should not be required to trigger or manage normal workflow states.

Open points: Define which operational actions belong inside core WMS services and which may be exposed through external API-based tools.

CMS-WM-002 - External request model independent of internal workflow implementation

Level: Hard requirement

Definition priority: P0 - Blocking design decision

Scope / likely owner: WMS core

Supporting groups: PPD, P&R, Core Software, Physics Coordination, RECO

Requirement: External request systems and users must not need to choose internal workflow orchestration types. The WMS should map external workflow specifications onto the internal graph representation best suited for execution, recovery, monitoring, and data handling.

Open points: Define the practical workflow graph model and how much of it, if any, is exposed to requesters.

CMS-WM-003 - Flexible multi-step and multi-output workflow support

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS core

Supporting groups: Tier-0, Physics Coordination, Core Software, RECO

Requirement: The WMS must support flexible workflow structures with multiple processing steps, multiple output datasets from the same workflow or application step, and workflow branches needed for multi-trigger, multi-reconstruction, PromptReco, NANO AOD, or hybrid execution use cases.

Open points: Clarify which multi-output or multi-branch cases should remain within a single workflow graph and which, if any, should be represented as separate workflows. Join-style dependencies are covered separately in CMS-WM-004.

CMS-WM-004 - Join-style workflow dependencies

Level: Hard requirement

Definition priority: P0 - Blocking design decision

Scope / likely owner: WMS core

Supporting groups: Core Software, P&R, PPD, Physics Coordination

Requirement: The WMS must support workflow graph patterns where an application step can depend on outputs or products from multiple upstream steps. These join-style dependencies should allow workflows to avoid creating heavy intermediate data tiers or carrying forward content that is only needed later in the processing chain.

Open points: Define how join dependencies are represented in the workflow graph, how data dependencies are tracked across upstream branches, and how recovery, retry, provenance, and output registration work when a downstream step depends on multiple predecessors.

CMS-WM-005 - Request validation before large-scale execution

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: PPD, P&R, GEN, RECO, User Private Workflows

Requirement: Requests must be validated before large-scale execution. The WMS, together with stakeholder-owned preparation systems when needed, must catch invalid, inconsistent, unsupported, or policy-violating requests before they consume significant resources.

Open points: Define validation responsibilities between WMS, PPD/PdmV, GEN tools, RECO scouting, User Analysis tooling, Core Software, and policy owners.

CMS-WM-006 - Dry-run, preflight, and controlled validation modes

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS core

Supporting groups: User Private Workflows, P&R

Requirement: The system must provide preflight validation modes for user workflows, including the ability to test an application with a single grid job and receive a debugging sandbox if it fails. For WMS development and central-production workflows, dry-run modes should allow changes to be tested against real metadata without creating payload jobs where feasible.

Open points: Define what side effects are allowed during dry-run and what is meant by dry-run across the full system. To consolidate whether it is a hard requirement for user workflows, while being phaseable for full central-production and development dry-run coverage.

CMS-WM-007 - Production-like replay, clone, and controlled test/backfill workflows

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS core

Supporting groups: P&R, Tier-0

Requirement: The WMS must provide controlled mechanisms to clone workflows, override selected parameters, and execute replay, test, validation, or backfill workflows in isolated or explicitly selected infrastructures.

Open points: Define how replay/test/backfill execution avoids unintended production side effects and how production-like replay is isolated from normal production processing.

Notes: Merged generic P&R safe replay and Tier-0 replay requirements.

2. Priority, quotas, and resource policy

Section categorization matrix (requirement IDs)

Level \ Definition priority	P0	P1	P2	Total
Hard requirement	CMS-WM-008	CMS-WM-009, CMS-WM-010	—	3
Soft requirement	—	—	—	0
Open decision	—	—	—	0
Total	1	2	0	3

CMS-WM-008 - Unambiguous priority model

Level: Hard requirement

Definition priority: P0 - Blocking design decision

Scope / likely owner: WMS interface / dependency

Supporting groups: PPD, Tier-0, Submission Infrastructure, Physics Coordination

Requirement: The future WMS must expose a clear priority model across Tier-0, central production, user/private workflows, analysis-like activities, calibration workflows, validation workflows, and other major CMS activities.

Open points: Decide whether priorities are absolute, fair-share based, or optimized for throughput and resource utilization.

CMS-WM-009 - Resource partitioning, quotas, and fair-share enforcement

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: PPD, Submission Infrastructure, Tier-0, User Private Workflows, RECO, GEN

Requirement: The system must support robust resource partitioning, quotas, and fair-share enforcement across groups and activities. This includes broad activity classes such as Tier-0, MC/data production, user analysis, as well as scarce or specialized resources such as GPUs, high-memory slots, long-walltime resources, and specialized queues.

Open points: Define the fair-share model, enforcement layer, quota semantics, and whether subgroup shares are required or optional.

Notes: Merged resource partitioning, multi-layer scarce-resource fair share, and Tier-0 global fair-share enforcement requirements.

CMS-WM-010 - Actionable policy and resource-block feedback

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS interface / dependency

Supporting groups: PPD, Physics Coordination, RECO, User Private Workflows

Requirement: Requesters must receive understandable feedback when workflows are blocked, deprioritized, resource-constrained, or unable to run with the requested policy, data, or hardware constraints.

Open points: Define whether feedback is delivered through workflow status, job exit codes, notifications, web UI, client output, or a combination.

3. Resource characterization and heterogeneous execution

Section categorization matrix (requirement IDs)

Level \ Definition priority	P0	P1	P2	Total
Hard requirement	—	CMS-WM-011, CMS-WM-015, CMS-WM-016, CMS-WM-017	—	4
Soft requirement	CMS-WM-013	CMS-WM-012, CMS-WM-014	—	3
Open decision	—	—	—	0
Total	1	6	0	7

CMS-WM-011 - Resource scouting and characterization against declared hardware targets

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: PPD, RECO, GEN, Tier-0, Core Software

Requirement: The WMS must infer or validate CPU, memory, time/event, output-size, GPU memory, and specialized-resource requirements through scouting jobs or benchmark workflows. Scouting must validate requirements for declared or supported hardware targets rather than blindly scanning all possible hardware configurations.

Open points: Define which metrics scouting must provide, when scouting is mandatory, and how the results are fed back to request preparation and scheduling.

Notes: Merged automated resource inference and user-selected hardware target scouting requirements.

CMS-WM-012 - Resource catalog and requirement pre-validation

Level: Soft requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: RECO, GEN, Submission Infrastructure, PPD/PdmV

Requirement: Workflow creators should have access to a CMS-pool resource catalog or equivalent service so that proposed resource requirements can be checked against available hardware and site capabilities before execution.

Open points: Define what the resource catalog contains and how recommendations or blocks are communicated to workflow preparers.

Notes: Particularly important for heterogeneous workflows; phaseable more generally.

CMS-WM-013 - Heterogeneous architecture and resource-target specification

Level: Soft requirement

Definition priority: P0 - Blocking design decision

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: RECO, GEN, Core Software, Submission Infrastructure, Physics Coordination

Requirement: The system should support heterogeneous architecture requirements, including CPU architecture, OS/runtime constraints, GPU availability, GPU properties, and other accelerators when needed. Workflows should be able to specify a list of valid targets rather than a single fixed target when possible.

Open points: Determine whether workflows or applications explicitly declare required and allowed target architectures/resources, whether the WMS infers compatibility from application metadata, scouting, artifact catalogs, or runtime validation, or whether CMS adopts a hybrid model. See also OD-012.

CMS-WM-014 - GPU-optional execution and offload characterization

Level: Soft requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: RECO, User Private Workflows, Submission Infrastructure

Requirement: The system should support a mode beyond GPU-required and GPU-forbidden: GPU-optional execution. For workflows where GPU offloading is beneficial but not strictly required, the WMS should be able to express, characterize, and schedule the workflow without compromising correctness.

Open points: Define whether GPU-enabled workflows must always have a CPU fallback, and which benchmark suite characterizes CPU/memory changes due to offloading.

Notes: Merged optional GPU support and GPU impact benchmarking. Hard for relevant RECO/GPU workflows, phaseable globally.

CMS-WM-015 - High-memory and non-standard resource profiles

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: Tier-0, Submission Infrastructure, GEN, RECO

Requirement: The system must support workflows requiring high memory or non-standard resource profiles beyond default resource envelopes. Resource requests must be based on measured or validated needs rather than limitations of default site or VO-card assumptions.

Open points: Define validation of resource requests against available grid resources and failure feedback when no suitable resources exist.

CMS-WM-016 - Dynamic adjustment of task/job requirements before retry

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: Tier-0, P&R, Submission Infrastructure, User Private Workflows

Requirement: Authorized users or operators must be able to adjust selected task/job requirements before retry. This includes resource requirements, target sites, memory/runtime escalation, site restrictions, and possibly application or configuration parameters.

Open points: Define which parameters can be changed at job, task, or workflow level, who is authorized to change them, and how historical changes are tracked.

CMS-WM-017 - Architecture-compatible runtime artifact and environment selection

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: GEN, RECO, Core Software, Submission Infrastructure

Requirement: The workflow execution model must support architecture-compatible runtime artifact discovery and selection when payloads depend on architecture-specific artifacts or controlled runtime environments, such as MadGraph gridpacks. Nested containers, architecture-bound gridpacks, legacy runtime environments, and non-default execution contexts are examples, not prescribed solutions.

Open points: Decide whether artifact selection is WMS-owned, application-wrapper-owned, or handled by external GEN/Core tooling; define the supported runtime-environment patterns; and align artifact selection with the application/resource-target contract in OD-012.

Notes: Hard where architecture-specific artifacts remain relevant, otherwise an open decision.

4. Data-aware execution and output integrity

Section categorization matrix (requirement IDs)

Level \ Definition priority	P0	P1	P2	Total
Hard requirement	CMS-WM-018, CMS-WM-026	CMS-WM-019, CMS-WM-020, CMS-WM-021, CMS-WM-022, CMS-WM-023, CMS-WM-024, CMS-WM-025	—	9
Soft requirement	—	—	—	0
Open decision	—	—	—	0
Total	2	7	0	9

CMS-WM-018 - Tight WMS/DM integration without responsibility duplication

Level: Hard requirement

Definition priority: P0 - Blocking design decision

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: PPD, DM, Tier-0, P&R

Requirement: The WMS must be tightly integrated with data management, remaining aware of data placement, storage quotas, availability, data rules, and transfer state to make correct workflow decisions without duplicating DM/Rucio responsibilities.

Open points: Define whether WMS actively makes destination decisions or consumes decisions from DM/Rucio.

CMS-WM-019 - Rucio account/activity separation for WM-managed data

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS interface / dependency

Supporting groups: DM, Tier-0, User Private Workflows

Requirement: WM-managed data must use a Rucio account/activity model that provides clear separation, accountability, and data ownership across major activity classes.

Open points: Define the account/activity model for Tier-0, user workflows, production, validation, and other relevant activities.

CMS-WM-020 - Dynamic Rucio placement logic and rule-state reconciliation

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: DM, PPD, Tier-0

Requirement: The WMS must work with generic RSE expressions and DM-controlled placement attributes, remain aware of Rucio-driven repositioning, and reconcile effective rule state from the data-management source of truth when rules or expressions are modified externally.

Open points: Define the DID level for rule evaluation, reconciliation frequency, and whether rule-related policies are configuration- or plugin-driven.

Notes: Merged generic RSE expression and externally modified rule-state requirements.

CMS-WM-021 - Block-level data-rule lifecycle and cleanup

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: DM, Tier-0

Requirement: Disk rule cleanup must be manageable at block level for input and output data, with policies depending on processing state, output persistence, transfer completion, and whether output is confirmed on tape or another long-term destination.

Open points: Clarify whether DM can own cleanup policy directly given Rucio relationship limitations such as lack of native 1:N rule association.

CMS-WM-022 - Controlled processing of partially available data

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: DM, P&R, PPD

Requirement: The WMS must allow authorized operators or workflow owners to force progress when some files of a block or container are missing, with clear auditability and explicit state tracking.

Open points: Define who owns the decision to process partially available data and whether this conflicts with workflow-completion semantics. Requires explicit policy controls.

CMS-WM-023 - System-managed data locks for processing safety

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: DM

Requirement: WMS processing must rely on data availability guarantees created or validated by the system. Unreliable or unmanaged locks/rules should not be used as the basis for location or processing safety.

Open points: Define how this applies to user workflows, Tier-0 streamer input, externally repositioned data, and centrally managed production data.

CMS-WM-024 - Output stageout integrity, retry, cleanup, and publication handoff

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: DM, User Private Workflows, DBS/DAS

Requirement: Output stageout must support retries, checksum verification, cleanup of failed or partial transfers where possible, fallback behavior, and consistent handoff to publication/registration systems. User and production output policies may differ, but both must preserve data integrity and auditability.

Open points: Define the stageout mechanism, retry boundaries, publication granularity, and how partial publication or partial transfer states are detected, retried, or rolled back.

Notes: Merged explicit checksum verification with user-output stageout integrity concepts.

CMS-WM-025 - Workflow-to-data-management traceability

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: DM, P&R, Tier-0

Requirement: The WMS must provide operator-facing entry points to trace which data-management rules, DIDs, transfers, and replicas are associated with a given workflow, task, or output product. The implementation should not prescribe whether the association is stored in WMS, Rucio metadata, a cross-reference table, or monitoring indexes.

Open points: Define the query entry points needed by operators and which system stores the authoritative association.

CMS-WM-026 - Runtime-generated globally unique output identifiers

Level: Hard requirement

Definition priority: P0 - Blocking design decision

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: User Private Workflows, DM, DBS/DAS, P&R

Requirement: Output files must use runtime-generated globally unique identifiers to avoid deterministic naming clashes across retries and recovery, while preserving traceability from workflow, task, job, retry attempt, and output product to the final registered file.

Open points: Define the identifier format, traceability metadata, and how uniqueness interacts with DBS/Rucio registration and workflow recovery.

5. Tier-0 online/offline processing

Section categorization matrix (requirement IDs)

Level \ Definition priority	P0	P1	P2	Total
Hard requirement	CMS-WM-029, CMS-WM-030	CMS-WM-027, CMS-WM-028, CMS-WM-031, CMS-WM-033, CMS-WM-034	CMS-WM-032, CMS-WM-035	9
Soft requirement	—	—	—	0
Open decision	—	—	—	0
Total	2	5	2	9

CMS-WM-027 - Permanent Tier-0 availability

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS core

Supporting groups: Tier-0, P&R

Requirement: Tier-0 services must support permanent availability during data-taking. The deployment and operating model must reflect that Tier-0 is always running.

Open points: Define operational availability targets, failover strategy, maintenance model, and service-health indicators.

CMS-WM-028 - Tier-0 throughput readiness and WMS implications of HL-LHC intake

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: Tier-0

Requirement: The Tier-0 WMS architecture must be compatible with HL-LHC online-to-offline throughput, including the operational and scaling implications of expected P5-to-Tier-0 data rates. The 100 GB/s target is primarily an infrastructure/data-flow target, but it has WMS implications for availability, load balancing, job creation, memory usage, and monitoring.

Open points: Define which parts of the throughput requirement are WMS-owned versus storage/network/DAQ-owned, as the throughput target has a WMS/infrastructure ownership split.

CMS-WM-029 - Horizontally scalable Tier-0 processing with uniqueness guarantees

Level: Hard requirement

Definition priority: P0 - Blocking design decision

Scope / likely owner: WMS core

Supporting groups: Tier-0, P&R

Requirement: Tier-0 processing components must scale horizontally and support load balancing while preserving uniqueness guarantees for production processing. The system must process all production data safely and avoid duplicate processing of the same logical production unit unless explicitly configured for replay or validation.

Open points: Define the unit of uniqueness and the coordination model across processing instances.

Notes: Merged Tier-0 horizontal scalability and exact-once production processing requirements.

CMS-WM-030 - Online/Tier-0 input handshake and streamer-file semantics

Level: Hard requirement

Definition priority: P0 - Blocking design decision

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: Tier-0, DBS/DAS, Core Software

Requirement: Tier-0 must support the online/Tier-0 input boundary, including the Storage Manager handshake, streamer files that may enter processing before being known to offline bookkeeping systems, and streamer-file lumisection semantics. The system must support cases where one lumisection is split across multiple streamer files while output RAW files preserve complete lumisections per primary dataset where required.

Open points: Define handshake granularity, metadata exchanged, and the final Run 4 semantics for split lumisections and output RAW completeness.

Notes: Merged Storage Manager handshake, unindexed streamer input, and split streamer lumisection requirements.

CMS-WM-031 - PPD-driven primary dataset definition interface

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS interface / dependency

Supporting groups: Tier-0, PPD, Physics Coordination

Requirement: Primary datasets must be created based on trigger paths and dataset definitions provided by the authoritative PPD/trigger source. The WMS must consume this information through a well-defined interface.

Open points: Define the authoritative source, update cadence, validation rules, and WMS behavior when definitions change.

CMS-WM-032 - PromptReco timing and conditions-readiness controls

Level: Hard requirement

Definition priority: P2 - Detailed design / phased capability

Scope / likely owner: WMS interface / dependency

Supporting groups: Tier-0, PPD

Requirement: PromptReco delay must be configurable in Tier-0 configuration, with boundaries such as run number where appropriate. It would be useful to allow conditions readiness or AICa green-light information to delay or release processing beyond the fixed delay model.

Open points: Define whether the fixed delay can be overridden by conditions readiness and what external framework owns that decision.

Notes: Configurable fixed delay is hard, automatic conditions-readiness override remains soft/open.

CMS-WM-033 - Tier-0 remote read and remote write model

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: Tier-0, DM, Submission Infrastructure

Requirement: Tier-0 workflows must support remote reading from Tier-0 disk by WLCG processing resources, and jobs running away from CERN must write output remotely to CERN EOS or the approved Tier-0 destination. Tier-0 should not depend on arbitrary site disk for input availability.

Open points: Define Rucio/data rules protecting Tier-0 disk until transfer and processing are complete, and define the exact remote-write mechanism.

CMS-WM-034 - Retain Tier-0 external interface capabilities and permanent metadata access

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS interface / dependency

Supporting groups: Tier-0, Physics Coordination, P&R

Requirement: Interface capabilities between Tier-0 and external systems must be retained. Selected Tier-0 metadata, such as workflow metadata and reconstruction configuration for each primary dataset, must remain permanently accessible through supported APIs or equivalent access mechanisms.

Open points: Define external consumers, required metadata, read/write responsibilities, persistence policy, and authorization model.

CMS-WM-035 - Tier-0 calibration and DQM output interface capability

Level: Hard requirement

Definition priority: P2 - Detailed design / phased capability

Scope / likely owner: WMS interface / dependency

Supporting groups: Tier-0, PPD, Physics Coordination

Requirement: The WMS must be able to execute or interface with Tier-0-related calibration workflows and provide DQMIO or other required outputs needed by downstream data-quality systems, when those workflows are defined as WMS-executed activities.

Open points: Define which calibration and DQM workflows are created by external frameworks, which are executed by WMS, and which outputs the WMS must make available.

Notes: Hard where workflows are WMS-executed; external ownership remains open.

6. Recovery, retry, continuation, and sample topping-off

Section categorization matrix (requirement IDs)

Level \ Definition priority	P0	P1	P2	Total
Hard requirement	CMS-WM-036	CMS-WM-037, CMS-WM-038	—	3
Soft requirement	—	—	—	0
Open decision	—	—	—	0
Total	1	2	0	3

CMS-WM-036 - Workflow recovery, retry distinction, and remaining-work completion

Level: Hard requirement

Definition priority: P0 - Blocking design decision

Scope / likely owner: WMS core

Supporting groups: P&R, User Private Workflows, Physics Coordination, PPD

Requirement: The system must natively identify failed, incomplete, or not-yet-executed units of work within a workflow and support recovery of those units through a controlled recovery mechanism. The system must distinguish automatic job retry within an existing task identity from explicit workflow recovery, which may require parentage or recovery relationships for auditability and provenance.

Open points: Define the recovery entity and identity model. Discussions with the P&R team indicate a preference for a single recovery entity that attempts to recover all missing work in a given workflow; remaining details include whether recovery remains under the same workflow ID, creates a new workflow with parentage, and whether task-specific recovery is supported.

Notes: Merged workflow recovery, retry/recovery distinction, and remaining-work completion requirements.

CMS-WM-037 - Automatic retry and resubmission policies

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: User Private Workflows, P&R, Submission Infrastructure

Requirement: Automatic retry and resubmission policies must exist, including site restrictions, alternative-site retry, memory/runtime escalation thresholds, and smaller data chunks until the minimum atomic processing unit is reached where applicable.

Open points: Define which retry policies are mandatory, which can be configured by users or operators, and when retry attempts stop.

CMS-WM-038 - Sample topping-off and production continuation

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: Physics Coordination, GEN, P&R, DBS/DAS

Requirement: The system must support controlled topping-off of existing samples, allowing approved requests to write additional events to datasets that already exist when policy and bookkeeping rules permit it. The model must preserve consistent metadata, parentage, dataset identity, duplicate avoidance, random-seed uniqueness, and traceability.

Open points: Define how topping-off differs from current extensions, which parameters must remain fixed across production rounds, which parameters must change, such as initial lumisection or random-seed ranges, and which approval or validation rules allow additional output to be associated with an existing dataset.

7. Observability, debugging, and operational interfaces

Section categorization matrix (requirement IDs)

Level \ Definition priority	P0	P1	P2	Total
Hard requirement	—	CMS-WM-039, CMS-WM-040, CMS-WM-042	—	3
Soft requirement	—	—	CMS-WM-041	1
Open decision	—	—	—	0
Total	0	3	1	4

CMS-WM-039 - End-to-end observability and workflow/job monitoring

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS core

Supporting groups: PPD, P&R, User Private Workflows, Physics Coordination, Tier-0, GEN, Submission Infrastructure

Requirement: A unified observability model is required, with common monitoring APIs and interfaces exposing workflow, data, site, service, processing, and operational states. Metrics, logs, events, workflow/task timelines, and job-level summaries must be available at the appropriate granularity.

Open points: Define which metrics are near-real-time, which are aggregated, and which belong to WMS, Submission Infrastructure, payload execution, or external monitoring systems.

Notes: Merged unified monitoring, timeline, per-job metrics, and payload/pilot-level monitoring requirements.

CMS-WM-040 - Actionable failure reporting, self-service debugging, and requester feedback

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: PPD, P&R, Core Software, User Private Workflows, Physics Coordination, RECO

Requirement: The system must provide robust error reporting, clear error categories, easy log access, grid-job sandbox access where available, failure-debugging context, and actionable feedback when workflows are blocked by data, policy, hardware, configuration, or resource constraints. The system should reduce reliance on expert-only hero-mode support.

Open points: Define error taxonomy ownership, uniqueness guarantees across WMS/CMSSW/XRootD, and whether debugging relies on preserved files, reproducible sandboxes, clone-and-tweak execution, or combinations.

Notes: Merged requester feedback, no-hero-mode, and failure-debugging context requirements.

CMS-WM-041 - Operator and user interfaces, logbooking, and notifications

Level: Soft requirement

Definition priority: P2 - Detailed design / phased capability

Scope / likely owner: WMS interface / dependency

Supporting groups: P&R, Physics Coordination, User Private Workflows

Requirement: Operator- and user-facing interfaces should support common operational workflows: search, drill-down, one-click or guided actions, saved searches, notes/logbooking, report export, notifications, and subscriptions to relevant workflow or data events.

Open points: Identify which dashboard, logbook, notification, and webhook capabilities are mandatory for initial operations and which can be phased.

Notes: Merged operator UI, logbooking, and webhook/notification requirements. Some sub-capabilities may be hard where needed for operations.

CMS-WM-042 - Monitoring retention and archival policy

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS core

Supporting groups: P&R, User Private Workflows, Physics Coordination

Requirement: The system must define retention and archival policies for workflow, task, job, and monitoring information. Detailed operational information cannot be kept forever at full granularity, but enough history must remain available for operations, reproducibility, accounting, and legacy workflow inspection.

Open points: Define what is retained forever, archived, summarized, or dropped.

8. Metadata, data model, and external bookkeeping dependencies

Section categorization matrix (requirement IDs)

Level \ Definition priority	P0	P1	P2	Total
Hard requirement	CMS-WM-043, CMS-WM-046	CMS-WM-044, CMS-WM-045	—	4
Soft requirement	—	—	—	0
Open decision	—	—	—	0
Total	2	2	0	4

CMS-WM-043 - Luminosity Section atomic-unit compatibility

Level: Hard requirement

Definition priority: P0 - Blocking design decision

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: DBS/DAS, Core Software, Tier-0, User Private Workflows

Requirement: For workflows whose processing or bookkeeping semantics are based on luminosity sections, the WMS splitting, retry, recovery, dynamic-boundary, and bookkeeping model must respect the CMS decision that the Luminosity Section is the smallest atomic processing unit, including HL-LHC one-second lumisections where applicable.

Open points: Define how the atomic-unit model applies to MC, streamer files, CMSSW-based user workflows, DQM/harvesting-like workflows, non-CMSSW applications, and dynamic job resizing or safe-boundary behavior.

Notes: This is a design constraint for WMS splitting, retry, recovery, and bookkeeping.

CMS-WM-044 - Consume and propagate the CMS-agreed metadata model

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS interface / dependency

Supporting groups: DBS/DAS, GEN, Physics Coordination, DM, PPD

Requirement: The WMS must consume and propagate the CMS-agreed metadata model for data tiers, parentage, MC/data identity, workflow provenance, and output products. The WMS should not define the full data-tier metadata model itself, but it must preserve and expose the required metadata through its interfaces.

Open points: Define which metadata belongs in DBS, Rucio, WMS, McM or successor systems, GEN tools, or external bookkeeping services.

Notes: Metadata definition is external to WMS.

CMS-WM-045 - Scalable bookkeeping interfaces as a WMS dependency

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: External dependency / policy decision

Supporting groups: DBS/DAS, User Private Workflows, Tier-0

Requirement: The WMS must interact with DBS and related bookkeeping systems through interfaces that scale to HL-LHC insertion and query loads and respect supported transaction, bulk-insertion, and query models.

Open points: DBS internal partitioning, archival, insertion refactoring, and query scaling are DBS responsibilities, but WMS design must not assume unscalable behavior.

Notes: Implementation is mostly external; WMS owns the scalable interface dependency.

CMS-WM-046 - Compatibility with externally governed data and bookkeeping models

Level: Hard requirement

Definition priority: P0 - Blocking design decision

Scope / likely owner: WMS interface / dependency

Supporting groups: DBS/DAS, User Private Workflows, Physics Coordination, Core Software

Requirement: The WMS must remain compatible with CMS data-model and bookkeeping decisions that are governed outside the WMS, including MC luminosity-section conventions, user/production data-scope models, parentage semantics, dataset naming or aliasing, and sample-discovery metadata. The WMS should consume authoritative definitions from the responsible systems and avoid hard-coding assumptions that would prevent those models from evolving.

Open points: Resolve the external decisions on MC luminosity-section semantics, phys03/global or successor data-scope model, cross-scope parentage, dataset naming/aliasing, and sample-discovery ownership.

9. User/private workflows

Section categorization matrix (requirement IDs)

Level \ Definition priority	P0	P1	P2	Total
Hard requirement	CMS-WM-047	CMS-WM-048, CMS-WM-049, CMS-WM-050, CMS-WM-051, CMS-WM-053	—	6
Soft requirement	—	—	CMS-WM-052	1
Open decision	—	—	—	0
Total	1	5	1	7

CMS-WM-047 - User-facing behavior and configuration migration

Level: Hard requirement

Definition priority: P0 - Blocking design decision

Scope / likely owner: WMS interface / dependency

Supporting groups: User Private Workflows, Physics Coordination

Requirement: Current user-facing capabilities such as submit, status, resubmit, kill, report, and getlog must be deliberately migrated to clearly documented replacements. Configuration semantics affecting PSet, input dataset, output tag, splitting, site constraints, storage, routing, and data access must not be silently lost.

Open points: Decide whether to maintain a CRAB-compatible layer during migration, move directly to a new interface, or provide both during a transition period.

Notes: Merged former user behavior and configuration-semantics requirements.

CMS-WM-048 - Authentication, authorization, ownership, and operator roles

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS interface / dependency

Supporting groups: User Private Workflows, P&R

Requirement: The system must use modern CMS IAM/OIDC-style identities as the target model. Work must be owned/accounted to the submitting user identity. State-changing operations are restricted to the owner or operator/super-user roles, while monitoring can remain broadly readable to authenticated CMS users.

Open points: Define token management during workflow lifetime and confirm whether any group-level permission model is required. Current CRAB-side experience suggests group-level permissions may be unnecessary and operationally costly unless a concrete use case is established.

Notes: Merged authentication and task ownership requirements.

CMS-WM-049 - CMSSW and non-CMSSW payload provisioning for user workflows

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS interface / dependency

Supporting groups: User Private Workflows, Core Software, Physics Coordination

Requirement: User workflows must provision CMSSW environments when needed, including SCRAM_ARCH and CVMFS access, but should not be restricted to CMSSW applications. The system must support arbitrary user applications within defined policy and security boundaries.

Open points: Define which non-CMSSW payload classes are in scope for the first deployment.

Notes: Hard for future user workflows, phaseable for central production.

CMS-WM-050 - User data access through local reads and configurable remote fallback

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS interface / dependency

Supporting groups: User Private Workflows, Core Software, Physics Coordination

Requirement: CMS data access for user workflows must support local reads and configurable remote fallback through AAA/XRootD or successor mechanisms. Users should be able to control whether workflows allow remote data access, while the WMS uses data and resource information to make scheduling decisions.

Open points: Define the future replacement for ignoreLocality-style behavior and how remote fallback is disabled or enabled.

CMS-WM-051 - User splitting, lumi bookkeeping, and retry consistency

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS interface / dependency

Supporting groups: User Private Workflows, DBS/DAS, Core Software

Requirement: Existing splitting modes, including file-based, lumi-based, event-based, automatic wallclock-driven splitting, and, where still needed for DQM or harvesting-like workflows, run-based and dataset-based splitting, must be supported or replaced with equivalent functionality. Job retries must preserve task identity and strict lumi bookkeeping without losing or duplicating processed units.

Open points: Study usage of splitting modes, including run-based and dataset-based modes, and define how splitting may change across retries while preserving bookkeeping. Generic retry and recovery semantics are covered by CMS-WM-036 and CMS-WM-037.

CMS-WM-052 - Programmatic and web access for user workflow services

Level: Soft requirement

Definition priority: P2 - Detailed design / phased capability

Scope / likely owner: WMS interface / dependency

Supporting groups: User Private Workflows, Physics Coordination

Requirement: A Python client library, Jupyter-friendly interface, and self-service web dashboard would be valuable for programmatic submission, monitoring, reports, and user workflow inspection.

Open points: Decide which interfaces are required for the first deployment and which can be delivered later.

CMS-WM-053 - Private or semi-official MC production path

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS interface / dependency

Supporting groups: Physics Coordination, GEN, User Private Workflows, DM, DBS/DAS

Requirement: Users must have a well-defined path to run private or semi-official MC generation/simulation using production-like workflow logic under lower priority, quotas, validation rules, and namespace/publication boundaries. This path should be distinguished from ordinary private analysis by its use of production-like provenance, validation, data-management, and potential output reuse or promotion rules.

Open points: Define whether this complements or replaces private CRAB-like workflows, what approval and validation rules apply, what makes such production distinct from ordinary private analysis, and how outputs can be published, promoted, or reused. Requires policy boundaries.

10. GEN, simulation, and campaign evolution

Section categorization matrix (requirement IDs)

Level \ Definition priority	P0	P1	P2	Total
Hard requirement	—	CMS-WM-054, CMS-WM-055, CMS-WM-056, CMS-WM-057	—	4
Soft requirement	—	CMS-WM-058	CMS-WM-059	2
Open decision	—	—	—	0
Total	0	5	1	6

CMS-WM-054 - GEN validation pipeline interface, reviewer visibility, and automation integration

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS interface / dependency

Supporting groups: GEN, Physics Coordination, PPD/PdmV

Requirement: The WMS must provide stable interfaces to the GEN request-preparation and validation pipeline, consuming approved workflow inputs and exposing the metrics needed by GEN reviewers before production approval without hard-coding fragile dependencies on specific external services. Approved cards/fragments, CI or validation outputs, REANA results, CVMFS gridpack placement, McM or successor information, and related artifacts should be exchanged through agreed interfaces owned with the relevant GEN and production-preparation systems.

Open points: Define which checks remain human approvals, which are automated, which information is provided by WMS versus GEN/REANA/McM or successor tools, and which service-interface contracts are needed so external service ownership remains clear.

Notes: Merged GEN validation, reviewer visibility, and automation-pipeline integration requirements. Validation ownership is shared with or external to WMS.

CMS-WM-055 - GEN-level provenance and discoverability interfaces

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS interface / dependency

Supporting groups: GEN, DBS/DAS, Physics Coordination

Requirement: GEN-level provenance, including cards, fragments, model files, gridpacks, generator configuration, internal cross sections, validation outputs, and links to approved samples, must be stored or exposed through the appropriate CMS systems and remain traceable from WMS-executed production.

Open points: Define what is already covered by GEN tools or McM/successor systems and what must be propagated through WMS metadata.

CMS-WM-056 - Low-filter-efficiency MC production support

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: Physics Coordination, DBS/DAS, Core Software, GEN

Requirement: The central production model must support MC production with very small filter efficiency, avoiding pathological splitting/bookkeeping behavior and supporting target-statistics or maximum-time approaches where appropriate.

Open points: Define the right abstraction, how MC luminosity-section conventions affect the solution, and whether changes are needed in splitting, bookkeeping, or workflow orchestration.

CMS-WM-057 - Premixing-library bottleneck mitigation

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: Physics Coordination, DM, RECO, Simulation, Submission Infrastructure

Requirement: The WMS must support the computing model chosen to mitigate the HL-LHC premixing-library bottleneck, including continuous data-locality lookup, data-placement coordination, alternative staging, or alternative pileup-mixing strategies if approved.

Open points: Define which alternatives are technically and policy-wise acceptable and which aspects are WMS-owned.

CMS-WM-058 - Multi-trigger, reconstruction-variant, and incremental campaign support

Level: Soft requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: Physics Coordination, Core Software, RECO, PPD

Requirement: The system should support campaign models where multiple trigger menus or trigger/reconstruction variants can be produced from the same MC sample, and should support continuous or incremental campaign production where approved by CMS policy.

Open points: Define whether variants are represented as multiple outputs, multiple steps, branches, separate workflows, or separate campaigns, and what metadata model is required.

Notes: May become hard after policy/design agreement.

CMS-WM-059 - Alternative simulation workflow execution after CMS validation

Level: Soft requirement

Definition priority: P2 - Detailed design / phased capability

Scope / likely owner: WMS interface / dependency

Supporting groups: Physics Coordination, GEN, Core Software

Requirement: The WMS must be able to represent and execute alternative simulation workflows, including FastSim/FlashSim-like workflows, once they are validated and approved by CMS policy and have the required metadata/bookkeeping model.

Open points: Define validation gates, metadata requirements, and whether the payload is cmsRun-based or not. Conditional on CMS validation/adoption of alternative simulation workflows.

11. CMSSW/application execution contract

Section categorization matrix (requirement IDs)

Level \ Definition priority	P0	P1	P2	Total
Hard requirement	CMS-WM-061	CMS-WM-062, CMS-WM-063	—	3
Soft requirement	—	—	CMS-WM-064	1
Open decision	CMS-WM-060, CMS-WM-065	—	—	2
Total	3	2	1	6

CMS-WM-060 - Required legacy CMSSW execution path

Level: Open decision

Definition priority: P0 - Blocking design decision

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: Core Software, PPD, Physics Coordination, User Private Workflows

Requirement: CMS must define which legacy CMSSW releases and campaigns remain required, and the WMS must provide a viable execution path for that agreed scope. Existing WMS-CMSSW communication may be preserved, translated, or supported through adapters where direct changes to old releases are infeasible.

Open points: Define which legacy campaigns must be supported and whether some use cases can be served through user/private workflow support rather than central production. Scope to be defined before converting this into a hard implementation requirement.

CMS-WM-061 - Stable CMSSW runtime-customization contract

Level: Hard requirement

Definition priority: P0 - Blocking design decision

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: Core Software, Tier-0, User Private Workflows

Requirement: The WMS and CMSSW must define a stable runtime-customization contract. The PSet passed to cmsRun must remain complete and compatible with the CMSSW release being used, while WMS customizations should be decoupled from CMSSW configuration internals through an extensible document or key-value interface where feasible.

Open points: Define valid customization keys/values, compatibility with old releases, and whether PSet template generation on worker nodes can be avoided.

Notes: Merged PSet integrity and customization-decoupling requirements.

CMS-WM-062 - Random-number seed management

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS interface / dependency

Supporting groups: Core Software, Physics Coordination, GEN

Requirement: Random-number seed management must be explicit and reproducible. Jobs within the same workflow must never reuse random seeds, and seed assignment must support recovery, continuation, and sample topping-off without duplication.

Open points: Decide whether the WMS centrally pre-creates and assigns all random seeds, and define the required reproducibility level: job-level reproducibility, workflow-level reproducibility, clone/replay reproducibility, and topping-off or continuation without seed duplication.

CMS-WM-063 - Structured job-report contract

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: Core Software, P&R, PPD, User Private Workflows

Requirement: The WMS and CMSSW must define a stable structured job-report contract containing the information required for monitoring, retry, bookkeeping, validation, data registration, and reproducibility. The contract should identify the minimum meaningful low-level cmsRun/job information to be preserved, while treating the file format as an implementation choice rather than a requirement by itself.

Open points: Decide whether migrating away from XML is worth the CMSSW effort, what information must be included in the contract, and which existing job-report fields are required, obsolete, ambiguous, or better represented elsewhere.

CMS-WM-064 - Safe dynamic job boundaries

Level: Soft requirement

Definition priority: P2 - Detailed design / phased capability

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: Core Software, Physics Coordination, User Private Workflows

Requirement: The WMS should eventually support dynamic job length, such as stopping early or fetching more work, but only at safe processing boundaries that preserve atomic-unit semantics and allow correct reporting of completed and incomplete work.

Open points: Define whether CMSSW can guarantee stopping only at safe boundaries, how unfinished data is represented, and whether CPU-efficiency or feasibility studies are needed for dynamic boundaries, especially for many-core jobs.

Notes: Open decision on feasibility and ownership.

CMS-WM-065 - CPU-core usage and enforcement contract

Level: Open decision

Definition priority: P0 - Blocking design decision

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: Core Software, Submission Infrastructure, RECO

Requirement: CMS must define whether CPU-core limits are only communicated to applications such as CMSSW or actively enforced by the execution environment, for example through cgroups or equivalent resource controls. The decision must account for arbitrary non-CMSSW applications as well as CMSSW payloads.

Open points: Decide ownership and enforcement layer before converting this into a hard WMS implementation requirement. Define whether enforcement is soft, such as monitoring, logging, graceful termination, and preserved diagnostics, or hard, such as immediate eviction with possible loss of log information.

12. Deployment, development, scalability, and migration

Section categorization matrix (requirement IDs)

Level \ Definition priority	P0	P1	P2	Total
Hard requirement	—	CMS-WM-066, CMS-WM-067, CMS-WM-068, CMS-WM-069	—	4
Soft requirement	—	—	—	0
Open decision	—	—	—	0
Total	0	4	0	4

CMS-WM-066 - Versioned, stateless, and reproducible deployment model

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS core

Supporting groups: P&R, Tier-0, DM

Requirement: WMS configuration and releases must be centralized, version-controlled, reproducible, and deployable through health checks, integration tests, staged rollout, and rollback. Persistent database and filesystem state should be disentangled from application deployment where feasible, especially for Tier-0.

Open points: Define which components require stateless deployment and how upgrades proceed against populated and fresh databases.

Notes: Merged centralized deployment and stateless deployment requirements.

CMS-WM-067 - Contributor, testbed, and integration environment

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS core

Supporting groups: P&R, DM, Tier-0, User Private Workflows

Requirement: A smooth contributor and validation path is required, including development setups, sample workflows, mocks where possible, tests, contribution guidelines, and integration/testbed environments capable of exercising realistic WMS-DM-bookkeeping-execution flows safely.

Open points: Define which parts are WMS-owned and which require external integration environments such as safe Rucio/DM namespaces or production-data migration mechanisms.

CMS-WM-068 - Validated horizontal scalability and capacity targets

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: Co-owned WMS + external systems

Supporting groups: P&R, Tier-0, User Private Workflows, Submission Infrastructure

Requirement: The WMS architecture and its service/processing components must be horizontally scalable to support HL-LHC workflow volumes, Tier-0 processing, and user-analysis capacity targets. Quantitative capacity projections must be validated against expected Run 4/HL-LHC needs and operational bottlenecks.

Open points: Confirm official targets, including concurrent jobs, completed jobs per day, availability, and whether design should use average or peak Run 3 values scaled by HL-LHC factors. Preserve collected CRAB-like activity statistics as input to the capacity-target discussion, even if they do not provide the full HL-LHC picture. Numerical targets remain open until validated.

CMS-WM-069 - Smooth migration, coexistence, and validation period

Level: Hard requirement

Definition priority: P1 - Early architecture requirement

Scope / likely owner: WMS core

Supporting groups: Physics Coordination, User Private Workflows, Core Software, P&R

Requirement: The transition must be smooth, with old and new systems coexisting for a defined period where needed. Users and operators must be able to validate representative workflows before final migration, and previous workflow information, configurations, and operational history should remain inspectable for a suitable period.

Open points: Define migration length, fallback procedures, validation criteria, old-data visibility, training, and onboarding material. Also define how much LS3, Run 3, LS2, Run 2, and earlier workflow and data metadata must remain available during transition and early Run 4, for how long, and through which archival or query interface. See also OD-013.

Notes: Merged user parallel commissioning and global smooth-transition requirements.

13. Design principles

DP-001 - Community alignment with CMS autonomy

Classification: Design principle

Statement: CMS should prefer reusable, community-aligned components where they satisfy CMS needs, while preserving enough autonomy to support CMS-specific workflows, policies, and operations.

14. Cross-cutting open decisions

OD-001 - Priority model

Decide whether priority is absolute, fair-share based, or optimized for throughput/resource utilization.

OD-002 - Workflow abstraction

Decide how much of the internal workflow graph is visible to requesters and external systems.

OD-003 - Recovery identity model

Decide same workflow ID versus new recovery workflow ID with parentage, and whole-workflow versus task-level recovery.

OD-004 - User analysis migration strategy

Decide whether to preserve a CRAB-compatible layer, move to a new interface, or provide both during transition.

OD-005 - DBS/Rucio user-data model

Resolve the user/production data model, including phys03/global semantics, parentage, publication, and promotion of private/semi-official outputs.

OD-006 - Tier-0 horizontal processing coordination

Define how Tier-0 scales horizontally while maintaining uniqueness guarantees for production processing.

OD-007 - MC luminosity-section semantics

Define the CMS-agreed convention for MC lumisections and its impact on splitting, DBS records, and bookkeeping.

OD-008 - Heterogeneous-resource policy

Define how to express hard constraints, preferences, optional GPU, architecture-compatible artifacts, and scarce-resource fair-share.

OD-009 - CMSSW/WMS execution contract

Define stable runtime customization, job-report contract, legacy adapters, safe dynamic boundaries, and CPU-core enforcement.

OD-010 - Monitoring retention

Define what workflow/job details are retained forever, archived, summarized, or dropped.

OD-011 - Dataset naming, aliases, and sample-discovery ownership

Define the authoritative ownership and interface expectations for dataset naming, aliases, sample discovery, and aggregate browsing, including which parts belong to DBS/DAS, McM or successor systems, GEN/Physics bookkeeping, and which information the WMS must consume or expose.

OD-012 - Application/resource-target contract

Decide whether workflows or applications explicitly declare required and allowed target architectures/resources, whether the WMS infers compatibility from application metadata, scouting, artifact catalogs, or runtime validation, or whether CMS adopts a hybrid model. Define how required targets, preferred targets, optional accelerators, artifact availability, fallback behavior, and invalid target/resource combinations are represented.

OD-013 - Legacy workflow and data-metadata retention / archival access model

Decide how much LS3, Run 3, LS2, Run 2, and earlier workflow and data metadata must remain available during transition and early Run 4, for how long, and through which access model. Define whether metadata must be migrated to an archival database, exposed through a supported web service/API, kept queryable through existing systems, or summarized for long-term inspection.

Appendix A. Versioning log

Version / source	Change summary
v0 source material	Initial source material. Stakeholder-divided requirement documents and supporting Tier-0, CRAB/User Analysis, and workshop feedback documents.
v1	First unified second-round draft, with 95 numbered requirements and preliminary hard/soft/open classification.
v2	Structural cleanup: merged overlapping requirements, generalized implementation-specific wording, added Scope / likely owner, moved out-of-scope items to dependencies/open decisions/design principles, removed current-system component terminology, converged output identity on runtime-generated global UIDs.
v3	Identifier, consistency, and definition-priority cleanup: added stakeholder acknowledgments; changed numbered requirement IDs from UR to CMS-WM; added a Definition priority field to each numbered CMS-WM requirement and introduced the P0/P1/P2 design-sequencing scale; normalized several Level and Scope fields; reframed CMS-WM-046 as a WMS-facing compatibility requirement; added OD-011 for dataset naming/aliases/sample-discovery ownership; and applied minor wording and criticality refinements while preserving the overall 68-requirement structure.
v4	Requirement-categorization summary update: added a global requirement categorization summary and compact section-level 2D matrices showing which CMS-WM requirements fall into each Level / Definition priority category. No requirement wording, numbering, level assignment, definition-priority assignment, or scope classification was changed.
v5 (current document)	Added CMS-WM-004 for join-style workflow dependencies from stakeholder feedback; renumbered subsequent CMS-WM requirements; integrated Alan Malta reviewer feedback on workflow graph modeling, preflight validation, resource-policy priority, recovery, topping-off, lumi-boundary semantics, debugging sandboxes, user splitting, GEN interfaces, random seeds, structured job reports, dynamic boundaries, CPU enforcement, and scalability statistics; added OD-012 and OD-013; updated categorization summaries, P0 list, appendices, and table of contents; and produced a clean copy with reviewer comments removed.

Appendix B. v1-to-v5 treatment map

This appendix records how requirements from v1 were treated in v5. It is intended as an audit aid, not as a replacement for the requirement text above.

Former requirements	v1	v5 treatment	Notes
CMS-WM-001		CMS-WM-001	Kept and generalized wording to avoid current-system implementation names.
Post-v3 stakeholder feedback		CMS-WM-004	Added as join-style workflow dependencies; related to CMS-WM-003 but separated because it constrains the workflow graph model.
CMS-WM-002		CMS-WM-002	Kept and generalized as an external request model independent of internal implementation.
CMS-WM-003		CMS-WM-003	Kept and generalized as flexible multi-step and multi-output workflow support.
CMS-WM-004		CMS-WM-005	Kept separate from resource scouting; request validation is broader.
CMS-WM-005		CMS-WM-006	Kept and clarified importance for user workflows and phaseability elsewhere.
CMS-WM-006, CMS-WM-032		CMS-WM-007	Merged and promoted to hard operational requirement for replay/clone/test/backfill.
CMS-WM-007		CMS-WM-008	Kept separate from fair-share enforcement; priority semantics are distinct.
CMS-WM-008, CMS-WM-009, CMS-WM-010		CMS-WM-009	Merged into resource partitioning, quotas, and fair-share enforcement.

Former requirements	v1	v5 treatment	Notes
CMS-WM-011, CMS-WM-050, CMS-WM-051		CMS-WM-040	Merged into actionable failure reporting, self-service debugging, and requester feedback.
CMS-WM-012, CMS-WM-013		CMS-WM-011	Merged into resource scouting and characterization against declared hardware targets.
CMS-WM-014		CMS-WM-013	Kept and aligned toward soft requirement language for heterogeneous architecture support.
CMS-WM-015, CMS-WM-016		CMS-WM-014	Merged optional GPU mode and GPU offload characterization.
CMS-WM-017		CMS-WM-015	Generalized to high-memory and non-standard resource profiles.
CMS-WM-018		CMS-WM-016	Kept and generalized to dynamic adjustment of task/job requirements.
CMS-WM-019		CMS-WM-018	Kept as WMS/DM integration umbrella requirement.
CMS-WM-020		CMS-WM-019	Kept and marked as WMS interface/DM-owned policy.
CMS-WM-021, CMS-WM-024		CMS-WM-020	Merged into dynamic Rucio placement logic and rule-state reconciliation.
CMS-WM-022		CMS-WM-021	Kept and generalized as block-level data-rule lifecycle and cleanup.
CMS-WM-023		CMS-WM-022	Kept and clarified auditability and policy controls.
CMS-WM-025, part of CMS-WM-068		CMS-WM-024	Merged into output stageout integrity, retry, cleanup, and publication handoff.
CMS-WM-026		CMS-WM-025	Rewritten as workflow-to-data-management traceability; no longer prescribes Rucio metadata storage.
CMS-WM-027		CMS-WM-023	Kept as data locks and processing safety.
CMS-WM-028		CMS-WM-027	Kept.
CMS-WM-029		CMS-WM-028	Generalized as throughput readiness and WMS implications of HL-LHC intake.
CMS-WM-030, CMS-WM-031		CMS-WM-029	Merged; removed agent terminology.
CMS-WM-033, CMS-WM-034, CMS-WM-035		CMS-WM-030	Merged into online/Tier-0 input handshake and streamer-file semantics.
CMS-WM-036		CMS-WM-031	Kept as automatic retry and resubmission policies.
CMS-WM-037		CMS-WM-032	Kept and clarified hard fixed delay and optional conditions-readiness override.
CMS-WM-038		CMS-WM-033	Kept and generalized remote read/write model.
CMS-WM-039		CMS-WM-034	Rewritten to retain Tier-0 external interface capabilities and metadata access.
CMS-WM-040, CMS-WM-041, CMS-WM-042		CMS-WM-036	Merged into workflow recovery, retry distinction, and remaining-work completion.
CMS-WM-043		CMS-WM-037	Kept as automatic retry and resubmission policies.
CMS-WM-044		CMS-WM-038	Kept and reframed as a hard scalable-bookkeeping interface dependency.
CMS-WM-045		CMS-WM-026	Rewritten to converge on runtime-generated globally unique output identifiers.
CMS-WM-046, CMS-WM-047, CMS-WM-048, CMS-WM-049		CMS-WM-039	Merged into end-to-end observability and workflow/job monitoring.
CMS-WM-052, CMS-WM-053		CMS-WM-041	Merged into operator/user interfaces, logbooking, and notifications.

Former requirements	v1	v5 treatment	Notes
CMS-WM-054		CMS-WM-055, CMS-WM-062, CMS-WM-044	Kept and generalized as GEN-level provenance interfaces.
CMS-WM-055		CMS-WM-044	Reframed as WMS consuming/propagating agreed metadata model, not defining it.
CMS-WM-056		CMS-WM-045	Kept and reframed as a hard scalable-bookkeeping interface dependency.
CMS-WM-057		CMS-WM-043	Kept as LS atomic-unit compatibility.
CMS-WM-058		CMS-WM-046, OD-007	Moved to external data-model decisions and cross-cutting open decisions.
CMS-WM-059		CMS-WM-046, OD-005	Kept as an open decision until the required legacy scope is defined.
CMS-WM-060		CMS-WM-046	Moved to external data-model decisions; not a WMS-owned requirement.
CMS-WM-061		CMS-WM-046	Moved to external data-model decisions; not a WMS-owned requirement.
CMS-WM-062, CMS-WM-063		CMS-WM-047	Merged into user-facing behavior and configuration migration.
CMS-WM-064, CMS-WM-065		CMS-WM-048	Merged into authentication, authorization, ownership, and operator roles.
CMS-WM-066		CMS-WM-049	Kept and generalized.
CMS-WM-067		CMS-WM-050	Kept and generalized.
CMS-WM-068		CMS-WM-024, CMS-WM-026	Split into stageout/publication handoff and global output identifiers.
CMS-WM-069		CMS-WM-051	Kept as user splitting, lumi bookkeeping, and retry consistency.
CMS-WM-070		CMS-WM-052	Kept as soft programmatic/web access.
CMS-WM-071, CMS-WM-094		CMS-WM-069	Merged into smooth migration, coexistence, and validation period.
CMS-WM-072, CMS-WM-073, CMS-WM-074		CMS-WM-054	Merged into GEN validation pipeline interface and reviewer visibility.
CMS-WM-075		CMS-WM-055	Kept and generalized as GEN-level provenance interfaces.
CMS-WM-076, CMS-WM-077		CMS-WM-017	Merged and generalized as runtime artifact/environment selection.
CMS-WM-078		CMS-WM-056	Kept.
CMS-WM-079		CMS-WM-057	Kept and generalized.
CMS-WM-080		CMS-WM-058	Kept and generalized.
CMS-WM-081		CMS-WM-059	Generalized as alternative simulation workflow execution after CMS validation.
CMS-WM-082		CMS-WM-060	Kept as an open decision until the required legacy scope is defined.
CMS-WM-083, CMS-WM-084		CMS-WM-061	Merged into a stable CMSSW runtime-customization contract.
CMS-WM-085		CMS-WM-062	Kept.
CMS-WM-086		CMS-WM-063	Generalized as a structured job-report contract.
CMS-WM-087		CMS-WM-064	Kept as safe dynamic job boundaries.
CMS-WM-088		CMS-WM-065	Kept as an open CPU-core usage/enforcement contract.
CMS-WM-089, CMS-WM-090		CMS-WM-066	Merged into a versioned, stateless, reproducible deployment model.
CMS-WM-091		CMS-WM-067	Kept; external Rucio integration environment marked as dependency.
CMS-WM-092, CMS-WM-093		CMS-WM-068	Merged into validated horizontal scalability and capacity targets; generalized terminology.
CMS-WM-095		DP-001	Moved from numbered requirement to design principle.

Appendix C. Items declined as numbered WMS requirements

Former item	v5 treatment	Reason
Former CMS-WM-095	Moved to DP-001	Broad HEP alignment with CMS autonomy is valuable guidance but not a testable WMS capability.
Former CMS-WM-061	Moved to CMS-WM-046 and OD-011	Aggregate sample discovery and sample browsing are mainly DBS/DAS/CADI/Glance/McM/GEN-bookkeeping responsibilities.
Former CMS-WM-060	Moved to CMS-WM-046 and OD-011	Dataset aliases, naming, and versioning should be owned by metadata/bookkeeping/request systems; WMS should consume and propagate the chosen identifiers.
Former CMS-WM-059	Moved to CMS-WM-046 and OD-005	The phys03/global model is a DBS/User Analysis policy and architecture decision, not a WMS-owned implementation requirement.
Former CMS-WM-056	Reframed as CMS-WM-045	DBS insertion/query scalability is a DBS requirement; WMS needs scalable interfaces and must not rely on unscalable behavior.
Former CMS-WM-055	Reframed as CMS-WM-044	The data-tier metadata model is broader than WMS; WMS must consume and propagate the CMS-agreed model.
Former CMS-WM-058	Moved to CMS-WM-046 and OD-007	MC lumisection convention is a CMS data-model/Core/DBS/Physics decision; WMS must remain compatible with the chosen convention.
Former CMS-WM-081	Reframed as CMS-WM-059	FastSim/FlashSim validation/adoption is external policy; WMS must support alternative simulation workflows once approved.